SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	Shri Shanmuga Priya K
Register Number	192324035

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I Shri Shanmuga Priya K 192324035 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Introduction

A media company plans to migrate its video streaming platform to the cloud to support millions of users simultaneously with minimal buffering and high availability. Cloud computing provides scalable infrastructure, distributed storage, and intelligent content delivery, making it suitable for handling large streaming workloads efficiently.

Cloud Architecture Design

The architecture should be designed using multiple cloud services that work together to provide fast and reliable streaming. Video files can be stored in cloud object storage such as Amazon S3, Azure Blob Storage, or Google Cloud Storage. To reduce latency, a Content Delivery Network (CDN) caches video content at edge locations closer to users. When a user requests a video, the CDN delivers the cached copy instead of accessing the origin server every time.

A Load Balancer is placed in front of multiple application servers to distribute user requests evenly. The application servers run inside an Auto Scaling Group, allowing the cloud platform to automatically increase or decrease the number of servers depending on user traffic. User information and subscription details are stored in a managed database with read replicas, while Redis or Memcached is used for caching frequently accessed data.

Architecture Flow:

Users → CDN → Load Balancer → Auto Scaling Application Servers → Database & Cloud Storage

Role of Load Balancing and Auto Scaling

Load balancing ensures that incoming requests are shared equally among available servers, preventing any single server from becoming overloaded. If one server fails, the load balancer automatically redirects traffic to healthy servers, thereby improving availability.

Auto Scaling continuously monitors system metrics such as CPU usage, memory utilization, or the number of requests. During peak viewing hours, additional virtual machines are launched automatically. When demand decreases, unnecessary servers are removed, reducing infrastructure costs while maintaining performance.

Cost Optimization Strategies

Several techniques can be used to reduce operational costs without affecting performance.

- Use Reserved or Savings Plans for predictable workloads.
- Store rarely accessed videos in archival storage such as Amazon Glacier.
- Compress video files using efficient codecs to reduce storage requirements.
- Cache frequently viewed content using CDN to reduce bandwidth costs.
- Enable lifecycle policies to move old content automatically to cheaper storage.
- Continuously monitor cloud spending using cost management tools.

Advantages

- Supports millions of concurrent users.
- Reduces buffering and latency.
- Improves system availability.
- Automatically handles traffic spikes.
- Optimizes cloud resource utilization and cost.

Conclusion

By combining CDN, load balancing, caching, and auto scaling, the media company can build a highly scalable and cost-effective cloud architecture that delivers smooth video streaming with minimal buffering, even during periods of extremely high demand.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Introduction

A FinTech startup wants to deploy its applications across multiple cloud providers to improve availability, reliability, and avoid dependence on a single vendor. However, managing applications consistently across different cloud platforms can be challenging. Using containers and container orchestration provides a standardized and efficient deployment solution.

Proposed Solution Using Container Orchestration and Multi-Cloud

The application should first be packaged into Docker containers, which include the application code, libraries, and dependencies. This ensures that the application runs consistently on any cloud platform. To manage these containers efficiently, Kubernetes can be used as the container orchestration platform. Managed Kubernetes services such as Amazon EKS, Azure AKS, and Google Kubernetes Engine (GKE) can be deployed across multiple cloud providers.

Infrastructure can be created and managed using Terraform or Ansible, ensuring that the same configuration is deployed in every cloud environment. Container images should be stored in a secure container registry so that all cloud providers use the same application version.

Service Discovery and Scaling

In a Kubernetes environment, service discovery allows containers to communicate with each other without using fixed IP addresses. Kubernetes automatically assigns DNS names to services, making communication between application components simple and reliable.

Scaling is handled automatically based on workload. When user requests increase, Kubernetes launches additional container instances (pods). If traffic decreases, extra pods are removed to save resources. A Load Balancer distributes incoming requests evenly among the available pods, ensuring that no single container becomes overloaded.

The scaling process includes:

- Horizontal Pod Autoscaler (HPA) to increase or decrease pods based on CPU or memory usage.
- Cluster Autoscaler to add or remove worker nodes automatically.
- Load Balancer to distribute traffic across healthy containers.

Benefits of a Multi-Cloud Deployment Model

Deploying applications across multiple cloud providers offers several advantages. It improves system availability because if one cloud provider experiences an outage, the application continues to run on another provider. It also prevents vendor lock-in, giving the organization flexibility to choose services from different vendors. Multi-cloud deployment can improve performance by serving users from the nearest cloud region and provides better disaster recovery capabilities. Some major benefits are:

- High availability and fault tolerance.
- Business continuity during cloud failures.
- Better geographic coverage and lower latency.
- Flexibility to choose the best services from different providers.
- Reduced dependency on a single cloud vendor.

Risks and Challenges

Although multi-cloud provides many advantages, it also introduces certain challenges. Managing infrastructure across multiple cloud providers increases operational complexity. Different cloud platforms have different security policies, networking configurations, and pricing models, making administration more difficult. Data synchronization between clouds and maintaining consistent security policies also require careful planning.

Some common risks include:

- Increased management complexity.
- Higher networking and operational costs.
- Security and compliance challenges.
- Data synchronization issues.
- Monitoring multiple cloud environments.

Conclusion

By using Docker containers, Kubernetes orchestration, and a multi-cloud strategy, the FinTech startup can achieve consistent application deployment, automatic scaling, and high availability across different cloud providers. Although multi-cloud environments require careful management, they provide greater reliability, flexibility, and business continuity, making them an ideal solution for modern financial applications.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Introduction

An e-learning platform stores important data such as student records, course materials, assignments, examination results, and videos. Accidental deletion of this data can disrupt learning activities and cause significant losses. Therefore, the organization needs a reliable backup and disaster recovery (DR) strategy to ensure that data can be restored quickly with minimal downtime.

Cloud-Native Backup, Replication, and Versioning Solution

The organization should implement a cloud-native backup solution that automatically creates regular backups of all critical data. Services such as AWS Backup, Azure Backup, or Google Cloud Backup can be used to schedule daily and incremental backups.

To prevent permanent data loss from accidental deletion, object versioning should be enabled. Versioning stores multiple versions of the same file, allowing users to restore previous versions if a file is deleted or modified.

For disaster recovery, data should be replicated to another cloud region. If the primary region becomes unavailable due to a failure or disaster, the replicated data in the secondary region can be used to continue operations. Database snapshots and point-in-time recovery should also be enabled to restore databases to a specific point before the data loss occurred.

Achieving RTO and RPO

Two important disaster recovery metrics are Recovery Time Objective (RTO) and Recovery Point Objective (RPO).

- **Recovery Time Objective (RTO)** is the maximum acceptable time required to restore services after a failure. A low RTO can be achieved by maintaining standby servers, automating recovery procedures, and enabling automatic failover. This ensures that services are restored within a few minutes.
- **Recovery Point Objective (RPO)** is the maximum acceptable amount of data loss measured in time. It can be achieved through frequent backups, continuous database replication, and transaction log backups. This minimizes the amount of data lost if a failure occurs.

Steps to Ensure Business Continuity

To maintain continuous operation during failures, the organization should follow these steps:

- Schedule automatic daily and incremental backups.
- Enable versioning for cloud storage to recover deleted files.

- Replicate critical data to a secondary cloud region.
- Configure automatic failover for applications and databases.
- Perform regular disaster recovery testing to verify backup integrity.
- Continuously monitor backup jobs and system health.
- Maintain documented recovery procedures and train IT staff.

Advantages

A well-planned backup and disaster recovery strategy offers several benefits:

- Protects data from accidental deletion and system failures.
- Reduces downtime and ensures faster recovery.
- Minimizes data loss through continuous replication.
- Maintains uninterrupted access to learning resources.
- Supports business continuity and improves system reliability.

Conclusion

By combining cloud-native backups, cross-region replication, object versioning, and automated disaster recovery, the e-learning platform can effectively protect its data and quickly recover from accidental deletion or cloud failures. This approach ensures low RTO, low RPO, and uninterrupted educational services, providing a reliable and resilient cloud environment.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Introduction

A government agency manages both sensitive and non-sensitive information. Sensitive data, such as citizen records and confidential documents, must remain on-premises to comply with government regulations, while non-sensitive applications like public websites and information portals can be hosted in the public cloud. A hybrid cloud model combines the security of on-premises infrastructure with the scalability and flexibility of the public cloud.

Hybrid Cloud Architecture

In the proposed architecture, sensitive applications and databases remain in the agency's on-premises data center, while non-sensitive services are deployed in the public cloud. A secure connection, such as a Virtual Private Network (VPN) or dedicated links like AWS Direct Connect or Azure ExpressRoute, is established between the on-premises environment and the cloud to ensure safe communication.

Public users access cloud-hosted applications, while internal systems securely communicate with on-premises databases through the private connection. Firewalls and API Gateways are used to filter and protect network traffic between both environments.

Architecture Flow:

Users → Public Cloud Applications → Secure VPN/Direct Connect → On-Premises Database

Identity Management and Encryption

A strong identity management system is essential to ensure that only authorized users can access government resources. The agency should implement Identity and Access Management (IAM) with Role-Based Access Control (RBAC) so that users receive only the permissions required for their roles. Single Sign-On (SSO) can simplify user authentication, while Multi-Factor Authentication (MFA) adds an extra layer of security.

Data security should also be maintained through encryption. Data stored in databases and storage systems should be encrypted using AES-256 encryption, while data transferred between the cloud and on-premises environment should be protected using TLS/SSL protocols. Encryption keys should be securely managed using a Key Management Service (KMS).

Challenges in Managing Hybrid Cloud Environments

Although hybrid cloud offers flexibility and security, managing two different environments introduces several challenges.

Some of the major challenges include:

- Maintaining consistent security policies across cloud and on-premises systems.
- Ensuring compliance with government regulations and data privacy laws.
- Managing data synchronization between environments.
- Handling network latency during communication.
- Increased complexity in monitoring and administration.
- Higher infrastructure and maintenance costs.
- Integration of legacy systems with modern cloud services.

Advantages

The hybrid cloud model provides several important benefits:

- Protects sensitive government data by keeping it on-premises.
- Provides cloud scalability for public-facing applications.
- Supports regulatory and compliance requirements.
- Improves flexibility and resource utilization.
- Enhances business continuity through cloud backup and disaster recovery.

Conclusion

A hybrid cloud architecture enables government agencies to securely manage sensitive information while taking advantage of the scalability and cost-effectiveness of public cloud services. By implementing secure communication, strong identity management, encryption, and continuous monitoring, the agency can build a reliable, compliant, and efficient hybrid cloud environment.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Introduction

A Software as a Service (SaaS) provider must ensure that its application remains available to users at all times, even if an entire cloud region experiences a failure. To achieve this, the application should be deployed across multiple cloud regions with proper data replication, automatic failover, and continuous monitoring. This approach improves reliability, minimizes downtime, and ensures uninterrupted service.

Multi-Region Deployment Strategy

The application should be deployed in two or more cloud regions using an Active-Active or Active-Passive architecture.

- **Active-Active:** Both regions handle user traffic simultaneously. If one region fails, the other continues serving users without interruption.
- **Active-Passive:** One region acts as the primary site, while the secondary region remains on standby and becomes active only if the primary region fails.

A Global Load Balancer is placed in front of the application to direct user requests to the nearest healthy region. Each region contains application servers, databases, and storage, ensuring that services remain available even during regional outages.

Architecture Flow:

Users → Global Load Balancer → Region A / Region B → Application Servers → Replicated Database

Data Replication and Failover Mechanism

To protect data and maintain availability, the database should be replicated across regions.

- Synchronous replication is used within the same region to ensure that data is written to multiple database instances simultaneously, resulting in minimal or no data loss.
- Asynchronous replication is used between different regions to reduce latency while maintaining updated copies of the data.

If the primary region becomes unavailable, the cloud platform automatically detects the failure and redirects user traffic to the secondary region. This automatic failover ensures that users can continue accessing the application with little or no interruption.

Monitoring and Alerting Strategies

Continuous monitoring helps detect problems before they affect users. Cloud monitoring services such as Amazon CloudWatch, Azure Monitor, or Google Cloud Monitoring can be used to track application health and infrastructure performance.

Important metrics to monitor include:

- CPU and memory utilization.
- Network traffic and bandwidth usage.
- Application response time.
- Database performance and replication status.
- Server availability and error rates.
- Storage capacity and disk usage.

Alerts should be configured to notify administrators through email, SMS, or collaboration tools whenever critical thresholds are exceeded. Centralized logging tools and dashboards also help administrators quickly identify and resolve issues.

Advantages

A multi-region deployment offers several benefits:

- Ensures high availability and fault tolerance.
- Minimizes downtime during regional failures.
- Improves disaster recovery capabilities.
- Provides faster access for users in different geographic locations.
- Enhances customer satisfaction through uninterrupted service.
- Supports business continuity and reliable application performance.

Conclusion

Deploying the SaaS application across multiple cloud regions with data replication, automatic failover, load balancing, and continuous monitoring ensures that the application remains highly available even if one region goes down. This strategy provides fault tolerance, reduces downtime, protects critical data, and delivers a seamless experience to users, making it an ideal solution for modern cloud-based SaaS applications.